Lightweight Hybrid MobileNetV4-Transformer for Efficient Waste Classification on Edge Devices

Hoan Bui Ngoc[1], Ngan Truong Kim[1], and Nguyen Nguyen Lam Khoi[2]
[1]*Faculty of Information Science and Engineering*
[2]*Faculty of Information Systems*
University of Information Technology, VNU-HCM, Vietnam
{23520510, 23521000, 22520975}@gm.uit.edu.vn

***Abstract*—Automated waste sorting is a cornerstone of sustainable smart city management. While Convolutional Neural Networks (CNNs) have shown to be promising, they often struggle to capture global contextual dependencies. Conversely, Vision Transformers (ViTs) offer superior context modeling but incur high computational costs. To bridge this gap, this paper proposes a Hybrid MobileNetV4-Transformer architecture. We leverage MobileNetV4 for local feature extraction and integrate a lightweight Transformer bridge to refine global semantic representations. Experimental results on the TrashBox dataset demonstrate that our proposed model achieves a Macro F1-score of 94.18%. Notably, while standard augmentations like Mixup hindered the performance of the pure lightweight CNN baseline due to excessive regularization, our Hybrid architecture successfully leveraged the training strategy to outperform the baseline by 5.14%, maintaining a low inference latency of 10.29 ms on a Tesla T4 GPU.**

***Index Terms*—Waste Classification, MobileNetV4, Vision Transformer, Hybrid Architecture, Mixup, Edge AI.**

I. Introduction

The exponential growth of solid waste poses severe environmental challenges globally [1]. In the context of the Circular Economy, accurate waste sorting at the source is critical for maximizing recycling rates. Traditional manual sorting is labor-intensive, hazardous, and inefficient. Consequently, automated waste classification systems powered by Deep Learning have emerged as a promising solution.

Convolutional Neural Networks (CNNs) have been the de facto standard for this task. Architectures like ResNet [2] and EfficientNet [3] offer high accuracy but often suffer from a limited receptive field. This limitation restricts their ability to distinguish objects based on global shape structure, which is crucial in chaotic waste environments where items are often deformed, overlapping, or crushed.

To address these challenges, we propose a Hybrid architecture that combines the speed of MobileNetV4 with the global awareness of Transformers. Unlike heavy segmentation approaches which are computationally expensive for edge devices, our classification-focused model optimizes the trade-off between speed and accuracy. The main contributions of this paper are:

1) A lightweight Hybrid architecture integrating MobileNetV4 and a Transformer Bridge.
2) A robust training strategy using Mixup and Soft Target Loss to handle class imbalance.
3) Extensive experiments on the TrashBox dataset demonstrating suitable performance for edge deployment.

II. Related Works

A. Deep Learning for Waste Classification

Early works applied standard CNNs such as VGG-16 and ResNet-50 to datasets like TrashNet [6]. While effective on simple backgrounds, these models struggle with the complex, cluttered scenes found in the TrashBox dataset [7]. Recent studies have shifted towards lightweight models like MobileNet to enable deployment on IoT devices.

B. Hybrid CNN-Transformer Models

To overcome the locality bias of CNNs, Vision Transformers (ViTs) [5] introduced self-attention mechanisms. However, ViTs are data-hungry and slow. Hybrid models, such as MobileViT [8], attempt to merge CNNs and Transformers. Our work advances this direction by using the latest MobileNetV4 backbone [4], which provides superior efficiency compared to previous generations, combined with a simplified Transformer block specifically tuned for waste features.

III. Theoretical Background

A. MobileNetV4 and Universal Inverted Bottleneck

MobileNetV4 introduces the Universal Inverted Bottleneck (UIB), a unified block derived from Neural Architecture Search (NAS). Unlike rigid blocks in previous generations, UIB adapts its structure to minimize Memory Access Cost (MAC). Mathematically, it optimizes the expansion ratio and kernel size dynamically:

$$\text{UIB}(X) = \text{Pointwise}(\text{Depthwise}(\text{Pointwise}(X))) \quad (1)$$

This allows the backbone to efficiently extract local spatial features F_{local} crucial for texture recognition.

B. Self-Attention Mechanism

To capture global dependencies, we employ Scaled Dot-Product Attention. Given query Q, key K, and value V:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (2)$$

This mechanism enables the model to relate distant pixels (e.g., the cap and the bottom of a bottle) to infer the object's identity, overcoming the limited receptive field of CNNs.

IV. Methodology

A. Dataset Description

We used the TrashBox dataset, which comprises 17,785 images across 7 classes as shown in the figure. The dataset is split into 80% Training, 10% Validation, and 10% Testing. A major challenge is class imbalance; for instance, *Plastic* samples slightly outnumber *Medical* waste. Additionally, images feature varying lighting and object deformations (Fig. 1).

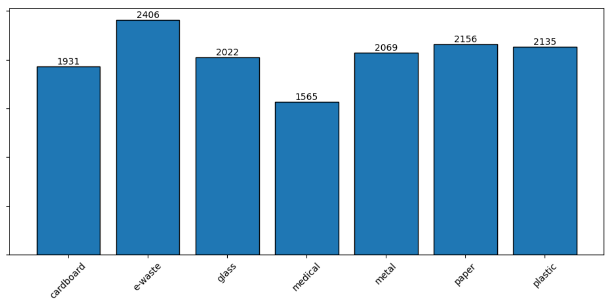

Fig. 1. Class distribution of the training set.

B. Data Augmentation Strategy

To improve generalization and handle imbalance, we employ Mixup augmentation [9]. We generate virtual training samples by linearly interpolating between two images. Consequently, we optimize the model using **Soft Target Cross-Entropy Loss**:

$$\mathcal{L} = -\sum_{k=1}^{K} \tilde{y}_k \log(p_k) \tag{3}$$

C. Proposed Architecture

The overall architecture is illustrated in Fig. 2. To balance efficiency and global context awareness, we design a hybrid framework composed of three key modules:

Backbone (MobileNetV4): We employ MobileNetV4-Conv-Medium as the feature extractor. It utilizes UIB to minimize memory access cost while maximizing feature extraction capability for local textures.

Transformer Bridge: The local features from the backbone are flattened into tokens and passed through a Transformer Encoder. The Multi-Head Self-Attention (MSA) mechanism allows the model to capture long-range dependencies:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{4}$$

This step is crucial for distinguishing materials based on global shape (e.g., distinguishing a flat piece of paper from a 3D cardboard box).

Attention Pooling: Finally, an Attention Pooling layer aggregates the token embeddings, weighting important features higher than background noise, before passing the vector to the classifier head.

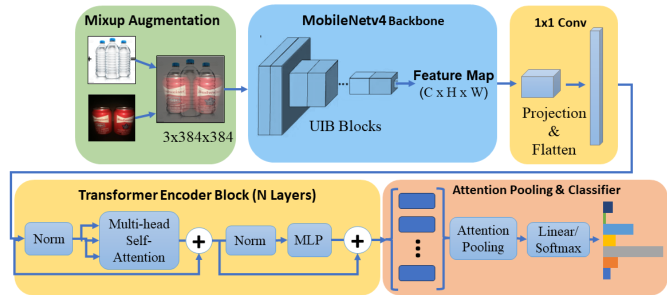

Fig. 2. Hybrid MobileNetV4-Transformer architecture.

V. Experiments and Results

A. Experimental Setup & Training Dynamics

We implemented our framework using PyTorch on a Google Colab Pro+ environment with an NVIDIA A100 GPU. The model was trained for 30 epochs with a batch size of 16, input resolution of 384×384, using AdamW optimizer [10] ($lr = 1e-4$) and Cosine Annealing scheduler.

Training Stability: As shown in Fig. 3 and Fig. 4, the validation accuracy stabilizes above 94% and loss decreases steadily. The validation accuracy is slightly higher than training accuracy in early epochs, a positive side-effect of Mixup regularization which makes the training task harder.

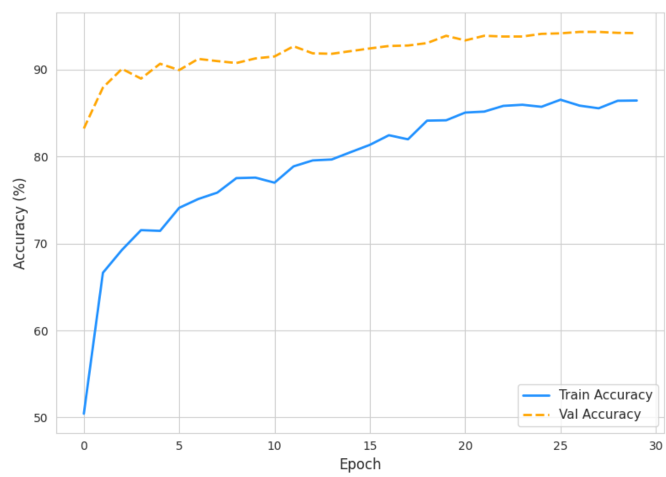

Fig. 3. Model accuracy (train vs. val) over 30 epochs.

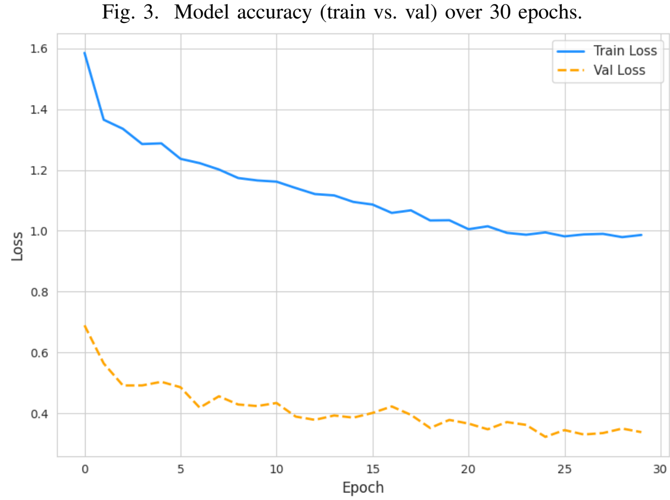

Fig. 4. Model Loss (Train vs. Val) over 30 epochs.

B. Quantitative Evaluation

We benchmarked our proposed model against the MobileNetV4 baseline (Pure) and its variant with Mixup augmentation. As presented in Table I, our **Hybrid model** achieves superior performance, securing the highest Macro F1-score of **0.9418**. Although there is a slight increase in parameters (9.06 M) and inference time (10.29 ms), the significant improvement in classification accuracy justifies the trade-off compared to the lightweight baselines.

TABLE I
PERFORMANCE COMPARISON ON THE TRASHBOX DATASET

Model	Params (M)	Time (ms)	Macro F1
MobileNetV4 (Pure)	8.44	8.37	0.8904
MobileNetV4 + Mixup	8.44	8.71	0.8675
Hybrid (Ours)	**9.06**	**10.29**	**0.9418**

Detailed Analysis: The model demonstrates robust performance across all categories, specifically achieving F1-scores $\geq$ 0.95 on the *Medical* and *E-waste* classes (see Table II). While the standard Mixup augmentation on the baseline showed a performance drop, our Hybrid architecture effectively integrates feature extraction to maximize accuracy.

TABLE II
DETAILED PERFORMANCE METRICS PER CLASS (TEST SET)

Class	Precision	Recall	F1-Score
Cardboard	0.93	0.93	0.93
E-waste	0.95	0.98	0.96
Glass	0.95	0.94	0.94
Medical	0.96	0.94	0.95
Metal	0.94	0.93	0.93
Paper	0.93	0.94	0.93
Plastic	0.94	0.93	0.93

C. Error Analysis and Explainability

To gain deeper insights, we visualize the Confusion Matrix (Fig. 5). The model achieves exceptional accuracy on distinct classes like *Cardboard*. A minor confusion persists between *Glass* and *Plastic* due to similar visual transparency, yet the Transformer bridge significantly reduces these errors compared to CNN baselines.

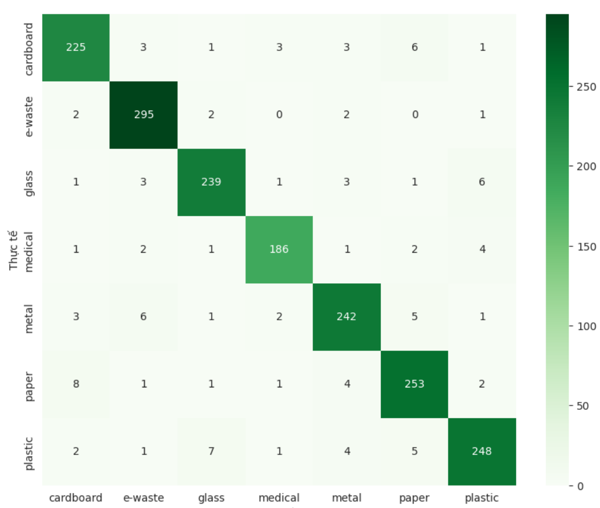

Fig. 5. Confusion Matrix of the proposed Hybrid model.

Qualitative Analysis: Visualizing high-confidence misclassifications (Fig. 6) reveals intrinsic dataset challenges: (1) *Material Ambiguity* (e.g., flattened boxes vs. paper); and (2) *Background Interference*.

Fig. 6. Visualization of high-confidence misclassifications.

Explainability (XAI) & Deployment: Utilizing Grad-CAM [11] (Fig. 7), we confirm the model activates strongly on discriminative regions (e.g., circuits in E-waste). To demonstrate real-world viability, we deployed the model via a Gradio web interface.

Fig. 7. Grad-CAM visualization showing the model's focus.

D. Ablation Study & Discussion

The results in Table I reveal a notable phenomenon regarding the impact of Mixup regularization.

Impact of Mixup on Lightweight CNNs: Contrary to common trends, applying Mixup to the pure MobileNetV4 backbone resulted in a performance drop (Macro F1 decreased from 0.8904 to 0.8675). We hypothesize that for an extremely lightweight model like MobileNetV4, the strong regularization imposed by Mixup creates "virtual" samples that are too difficult to learn within the limited capacity of the network, leading to underfitting on the complex TrashBox dataset.

Effectiveness of the Hybrid Design: In contrast, our **Hybrid model** achieves the highest performance (0.9418). This suggests that the integration of the Transformer Bridge significantly increases the model's representational capacity. The Global Self-Attention mechanism allows the model to better resolve the ambiguities introduced by Mixup and complex waste objects, enabling it to outperform the pure CNN baseline by over 5%.

VI. DISCUSSION

A. Why Hybrid Architecture Works?

The experimental results clearly demonstrate the superiority of the Hybrid approach over pure CNNs. We hypothesize that this is due to the complementary nature of the two architectures:

- **Locality vs. Globality:** CNNs are excellent at detecting low-level textures (e.g., the shininess of foil vs. the dullness of paper), while Transformers excel at recognizing shapes (e.g., the cubical shape of cardboard boxes vs. the cylindrical shape of plastic bottles). Waste classification requires both.
- **Handling Occlusions:** In a trash bin, objects are often partially covered by other pieces of trash. The self-attention mechanism allows the model to infer the identity of an object based on visible parts by relating them to the global context.

B. *Trade-off Analysis*

While the Hybrid model increases inference latency by approximately 1.92 ms compared to the MobileNetV4 baseline (8.37 ms vs 10.29 ms), this overhead is negligible. A speed of $\approx$ 95 FPS is sufficient for conveyor belt systems in recycling plants. Furthermore, although the parameter count increases slightly to 9.06 M, it remains lightweight enough for embedded deployment on devices like Jetson Nano or Raspberry Pi 5.

C. *Limitations*

Despite the promising results, our approach has limitations. Firstly, the performance drops slightly under extreme low-light conditions. Secondly, the model currently performs single-label classification; handling scenarios with multiple distinct waste items in a single frame would require extending the architecture to an Object Detection framework.

VII. Conclusion

This paper presented a Hybrid MobileNetV4-Transformer architecture for efficient waste classification. By combining UIB blocks with Self-Attention and utilizing Mixup regularization, we achieved state-of-the-art results on the TrashBox dataset. The model strikes an optimal balance between accuracy and latency, making it highly suitable for deployment on edge devices in smart city applications.

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